Timothy Stephens
Curriculum vitae

Computational Evolutionary Biologist

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Research profile

Computational evolutionary biologist working at the intersection of comparative genomics, evolutionary biology, and data science. My research investigates how genes with unknown function ("dark genes") originate, diversify, and acquire biological roles. I combine comparative genomics, phylogenomic analysis, sequence- and structure-based homology detection, and multi-omics approaches to investigate gene-family evolution and prioritize functional hypotheses across diverse biological systems.

My research spans coral, algal, amoebal, bacterial, and environmental microbial systems. I develop and apply computational workflows and analytical tools that address general problems in genomic discovery, from phylogenetic classification and genome annotation to metagenomic analysis and network-based interpretation. I have led collaborative computational research, mentored graduate and undergraduate researchers, and co-founded OceanOmics, translating biological research into environmental diagnostic technologies.

Research areas

- Evolutionary genomics and gene-family evolution
- Comparative genomics, phylogenomics, and phylogenetic methods
- Origins and evolution of novel and poorly characterized genes
- Genome annotation and computational protein characterization
- Metagenomics, multi-omics, and environmental genomics
- Reproducible bioinformatics workflows and research software

Academic appointments

2024–2026	**Research Associate**, Department of Biochemistry and Microbiology, Rutgers University Supervisor: Distinguished Professor Debashish Bhattacharya. Project focuses on coral conservation and algal evolution.
2019–2024	**Post Doctoral Associate**, Department of Biochemistry and Microbiology, Rutgers University Supervisor: Distinguished Professor Debashish Bhattacharya. Project focuses on using integrated multi-omics data to study plastid evolution and endosymbiosis in the photosynthetic amoeba *Paulinella*.

Selected professional and technical experience

2026–present	**Subject-Matter Expert, Bioinformatics & Genomics (Contract)**, Micro & Snorkel AI Develop research-level question-answer sets in bioinformatics, genomics, and multi-omics for frontier AI model training and evaluation, each with supporting references, quoted evidence, and a justification that the question has a single defensible answer.

2023–present	**Co-Founder, Chief Executive Officer and subsequently Chief Technology Officer**, OceanOmics, Inc. Co-founded an environmental biotechnology company and contributed to scientific and technical leadership of environmental health diagnostic platforms. Led or coordinated interdisciplinary project teams and contributed to securing more than $150,000 in company seed and voucher awards. Specify the distinction between company-awarded funding and personal academic grant support.

Education

2016–2019	**PhD**, Institute for Molecular Bioscience, The University of Queensland Supervisors: Professor Mark Ragan and Dr Cheong Xin Chan.
2015	**Honours**, Institute for Molecular Bioscience, The University of Queensland Supervisors: Professor Mark Ragan and Dr Cheong Xin Chan.
2012–2015	**Bachelor of Biotechnology**, Institute for Molecular Bioscience, The University of Queensland Major: Molecular Biotechnology and Innovation Management

Computational expertise and methods

Programming and research software: R package development and maintenance; Python; Perl; shell-based bioinformatics scripting; reproducible analysis and version-controlled code.

Comparative and evolutionary genomics: phylogenetic and phylogenomic analysis; phylogenetic-tree screening and classification; gene-family comparison; genome annotation; sequence-based homology searches; comparative analysis of divergent genes.

Metagenomics and multi-omics: Snakemake-based workflow development; metagenomic quality control, assembly, binning, gene prediction, functional annotation, and quantification; integration of transcriptomic, proteomic, and genomic evidence.

Network analysis and data structures: graph-based network analysis using Python graph-tool; connected-component extraction, graph annotation, and network visualization.

Computational infrastructure: Linux/HPC environments; Conda/Mamba environments; SLURM-oriented workflow execution; Snakemake; Docker, Singularity, and Podman containerization.

Research software and computational tools

PhySortR — Creator and maintainer of the R port of a package for screening and sorting large collections of phylogenetic trees, identifying exclusive and non-exclusive clades in traditional and extended Newick formats. Available through CRAN (version 1.0.9; package DOI: 10.32614/CRAN.package.PhySortR). Original method described in Stephens et al., *PeerJ* (2016).

Dinoflagellate Annotation Workflow — Developed a workflow for protein-coding gene prediction in dinoflagellate genomes, integrating transcriptomic, proteomic, and *ab initio* gene-prediction evidence. Includes adaptations for non-standard dinoflagellate splice-site biology. Source code.

naive ATLAS — Adapted and extended the ATLAS metagenomic workflow to support recovery and analysis of microbial genomes across prokaryotes, microbial eukaryotes,

viruses, and plasmids. Workflow includes quality control, assembly, binning, gene prediction, annotation, and quantification. Source code.

Network Analysis Scripts — Python utilities using graph-tool for graph construction, connected-component extraction, annotation, layout calculation, and visualization. Source code.

MAGS-CodeDev — Multi-agent graph-based software-development system that decomposes code projects into dependency graphs and coordinates code generation, testing, diagnosis, review, and iteration in isolated worktrees and containers. Source code.

AgentBox — Rootless Podman-based sandbox for running coding agents in project-scoped environments with explicit filesystem and network permissions. A software-infrastructure project demonstrating work in reproducible and constrained computational environments. Source code.

Research leadership and selected projects

Evolutionary origins of coral dark genes — NSF EDGE project
Senior personnel; computational analysis and manuscript leadership. Led computational analyses investigating the evolutionary origins of conserved dark-gene families in stony corals and contributed to the project's first research paper. The work found that approximately 82% of conserved coral dark-gene families were consistent with previously undetected homology to characterized gene families.
[Related publication: Stephens et al., Genome Biology and Evolution (2026).]

Computational enzyme discovery in Sargassum-associated microbial communities — Schmidt Sciences project
Senior personnel; computational discovery. Led computational discovery and prioritization of candidate enzyme families from microbial communities associated with Sargassum degradation, connecting comparative genomics with potential biomass-conversion applications.
[Related publication: Nandi et al., Applied and Environmental Microbiology (2026).]

Interdisciplinary environmental diagnostic development — OceanOmics
Co-founder; scientific/technical leadership. Led or coordinated interdisciplinary teams spanning biology, computer science, engineering, and synthetic biology in the development of environmental diagnostic platforms. Contributed to project development, technical problem-solving, funding acquisition, and progression of platforms toward proof of concept.

Grants and sponsored research

2026 New Jersey CSIT Clean Tech R&D Voucher Pilot Program, $39,760 USD
Funds awarded to our startup OceanOmics to utilize government facilities to aid development of our shellfish health monitoring toolkit.
Principal Investigators: **Timothy Stephens** (Rutgers University)

2024 New Jersey CSIT Clean Tech R&D Voucher Pilot Program, $39,860 USD
Funds awarded to our startup OceanOmics to utilize government facilities to aid development of our coral health monitoring toolkit.
Principal Investigators: **Timothy Stephens** (Rutgers University)

2024 New Jersey CSIT Clean Tech Seed Grant Program, $74,870 USD
Funds awarded to our startup OceanOmics to continue development of our coral health monitoring toolkit.
Principal Investigators: **Timothy Stephens** (Rutgers University)

2023 Rutgers TechAdvance Fund, $74,842 USD
Funds to continue development of our coral health monitoring toolkit.
Principal Investigators: Debashish Bhattacharya (Rutgers University)

2023 Climate Action Grant 2023, $10,000 USD
Funds for metabolomic and transcriptomic profiling of sex hormone dysregulation in spawning corals during thermal stress.
Principal Investigators: Debashish Bhattacharya (Rutgers University)

2023 Core Facility Utilization Application 2023, $5,000 USD
Funds for metabolomic profiling of the sex hormone cycle in spawning corals.
Principal Investigators: Debashish Bhattacharya (Rutgers University)

2022 Core Facility Utilization Application 2022, $5,000 USD
Funds for sequencing of coral microbiome samples.
Principal Investigators: Debashish Bhattacharya (Rutgers University)

2022 Center for Nutrition, Microbiome, and Health Small Grant FY-22, $2,000 USD
Characterizing the coral microbiome biogeography across colonies and reefs.
Principal Investigators: Debashish Bhattacharya (Rutgers University)

2018 UQ-FAPESP Strategic Research Fund SPRINT (2018/15159-9), $20,000 (~USD $14,000)
Integrated genomic approaches to understand stress tolerance in bioethanol-producing yeasts and coral reef symbionts
Principal Investigators: Cheong Xin Chan (UQ) and Jeferson Gross (State University of Sao Paulo)

Research mentoring and supervision

Served as a day-to-day research mentor to nine graduate students and five undergraduate researchers in computational biology, genomics, and related areas.

Graduate mentorship - Mentored three PhD students through completion.

- Each completed PhD mentee produced three to four first-author research papers during doctoral training.
- One former mentee is now an Assistant Professor.

Student-led research - Oversaw computational analyses for 16 student-led research papers.

- Twelve of these papers included me as a co-first or second author.

Mentoring included research-question development, computational study design, analytical troubleshooting, interpretation, reproducibility, manuscript preparation, and progressive transfer of project ownership.

Teaching experience

2020–2026 **Co-lecturer**
Rutgers University
Design and conducted lectures on de novo next-generation genome and transcriptome sequencing, metabolomics, proteomics, metagenomics, phylogenetics, and phylogenomics in a join undergraduate and postgraduate course titled "Fundamentals of Microbial Genomics". Duties include designing and marking student' exam questions and oral presentations.

2018 **Head Practical Tutor**
University of Queensland
Duties included designing and developing assessment material for the course and organising marking of assignments.

2016–2017 **Practical Tutor**
University of Queensland
Duties included assisting students with the completion of set questions and marking of assignments.

2013 **Science Mentor**
University of Queensland
Duties included organizing and running first year science student introduction and social engagement events designed to inform students about study/research opportunities available at The University of Queensland.

2013–2014 **Peer Assisted Study Session (PASS) Tutor**
University of Queensland
Duties included planning and leading multiple weekly tutorial sessions, each comprising 20-30+ students.

2012–2014 **CASPiE Tutor**
University of Queensland
Duties included guiding undergraduate chemistry students through an advanced set of research-focused practicals, with the aim of developing their critical and experimental thinking.

Professional development

2024 NSF I-Corps Regional Workshop, Rutgers University
2017 Research Commercialisation Workshop, University of Queensland
2017 UQ Idea Hub, University of Queensland

Awards and honours

2023 Selected to attend and present at the New Phytologist Next Generation Scientists Conference 2023, National University of Singapore
2021 2019 Dean's Award for Outstanding Higher Degree by Research Theses, University of Queensland
2018 Registration award for Society for Molecular Biology & Evolution (SMBE) 2018, SMBE
2017 Won best pitch prize at the UQ Idea Hub, University of Queensland
2016 Research Training Program (RTP) scholarship, University of Queensland
2016 Top poster prize at the IMB Research Higher Degree Student Symposium, University of Queensland
2012 Selected for the Advanced Study Program in Science (ASPinS; based on academic merit), University of Queensland
2012 Dean's Commendation for Academic Excellence (achieved a GPA over 6.6), University of Queensland
2011 Merit Scholarship for academic achievement, University of Queensland

Publications

2026 Van Etten J., Han S., Burns J. A., Lhee D., Willoughby A. C., **Stephens T. G.**, Chille E., Sleith R. S., Bhattacharya D., and Yoon H. S. Crawling under the radar: Two novel *Paulinella* species expand knowledge about the ecology and evolution of a primary plastid-containing amoeba lineage. *Journal of Phycology*, 2026. [DOI/URL] Featured by The New York Times

2026 dos Santos Ferreira M. C., Stephens T. G., and Brinsmade S. R. In *Staphylococcus aureus*, MbcS is a refunctionalized acyl-CoA synthetase that confers a fitness advantage during intra-species competition. *Journal of Bacteriology*, 208:e00185-26, 2026. [Preprint] [DOI/URL]

2026 Nandi S., **Stephens T. G.**, Garcia R., Sanchez-Garcia M. A., Roberson L. M., Avalos J., Chundawat S. P. S., and Bhattacharya D. Rafts of Change: Microbial and Functional Dynamics in Simulated Sargassum Strandings. *Applied and Environmental Microbiology*, 92:e02357-25, 2026. [DOI/URL] Project featured by Rutgers News

2026 **Stephens T. G.**, Kulczyk A. W, and Bhattacharya D. Cosmopolitan gene families with known functions are hotspots for the evolution of novel genes in stony corals. *GBE*, 18:evag072-4, 2026. [DOI/URL]

2026 *Nandi S., ***Stephens T. G.**, Chille E. E., Goyen S., Bay L. K., and Bhattacharya D. Metaproteome Analysis of Short-Term Thermal Stress in Three Sympatric Coral Species Reveals Divergent Host Responses. *Ecology and Evolution*, 16:e73275, 2026. [DOI/URL] Featured by the Rutgers Newsroom. *Co-first authorship

2026 Karimi K., Arroyo M., Chille E. E., **Stephens T. G.**, Barrett D., Pelechano V., Bhattacharya D., and Javanmard M. A portable, low-cost, point-of-care DNA amplification kit with impedance-based detection for decentralized antimicrobial resistance diagnostics. *Lab on a Chip*, 2026. [DOI/URL]

2026 Bhattacharya D., Nandi S., Chille E. E., Arroyo M., and **Stephens T. G.** The host coral bleaching response viewed through the lens of multi-omics. *BioEssays*, e70110, 2026. [DOI/URL]

2025 Nandi S., **Stephens T. G.**, Walsh K., Garcia R., Villalpando M. F., Sellares-Blasco R. I., Zubillaga A. L., Croquer A., Bhattacharya D. Shifts in the microbiome and virome are associated with stony coral tissue loss disease (SCTLD). *ISME Communications*, ycaf226, 2025. [Preprint] [DOI/URL] Featured by the Rutgers Climate and Energy Institute

2025 Goyen S., Chille E. E., **Stephens T. G.**, Nandi S., Bay L. K., Bhattacharya D. Green fluorescent proteins show divergent patterns among species and strains of *Porites* from the Great Barrier Reef. *Coral Reef*, 2025. [DOI/URL]

2025 Bhattacharya D., Van Etten J., Panayotakis G., McDermott T., **Stephens T. G.** Gene transfer drives community cooperation in geothermal habitats. *Trend in Microbiology*, 2025. [DOI/URL] Featured by Phys.org

2025 Chille E. E., **Stephens T. G.**, Nandi S., Jiang H., Gerdes M., Williamson O. M., Neufeld A., Montoya-Maya P., Bhattacharya D. Coral Restoration in the Omics Era: Development of Point-of-Care Tools for Monitoring Disease, Reproduction, and Thermal Stress. *BioEssays*, e70007, 2025. [DOI/URL]

2025 Van Etten J., **Stephens T. G.**, and Bhattacharya D. Genetic transfer in action: uncovering DNA flow in an extremophilic microbial community. *Environmental Microbiology*, 27(2):e70048, 2025. [DOI/URL]

2024 **Stephens T. G.**, Van Etten J., McDermott T., Christian W., Chaverra M., Gurney J., Lee Y., Kim H., Hyun Cho C., Chovancek E., Westhoff P., Otte A., Northen T. R., Bowen B. P., Louie K. B., Barry K., Grigoriev I. V., Mock T., Liu S., Miyagishima S., Yoshinaga M, Weber A. P.M., Yoon H. S., and Bhattacharya D. Temporal dynamics in a red alga dominated geothermal feature in Yellowstone National Park. *ISME Communications*, 4:01, 2024. [Preprint] [DOI/URL]

2024 Boyd J. M, Kaler K. R., Esquilin-Lebron K., Pall A., Campbell C. J., Foley M. E., Rios-Delgado G., Mustor E. M., **Stephens T. G.**, Bovermann H., Greco T. M., Cristea I. M., Carabetta V. J., Beavers W. N., Bhattacharya D., Skaar E. P., Shaw L. N., Stemmler T. L. Fpa (YlaN) is an iron(II) binding protein that functions to relieve Fur-mediated repression of gene expression in *Staphylococcus aureus*. *mBio*, 2024. [Preprint] [DOI/URL]

2024 Calatrava V.*, **Stephens T. G.***, Grossman A. R., and Bhattacharya D. What has *Paulinella* taught us about endosymbiont metabolic integration?. *Schwartzbach, S.D., Kroth, P.G., Obornik, M. (eds) Endosymbiotic Organelle Acquisition. Springer, Cham.*, 2024. [DOI/URL] *Co-first authorship

2024 Chille E. E., **Stephens T. G.**, Misri D., Strand E. L, Putnam H. M., and Bhattacharya D. Gene expression response under thermal stress in two Hawaiian corals is dominated by ploidy and genotype. *Ecology and Evolution*, 14:e70037, 2024. [DOI/URL]

2024 Gabr A., **Stephens T. G.**, Reinfelder J. R., Liau P., Calatrava V., Grossman A. R., and Bhattacharya D. Evidence of a putative CO2 delivery system to the chromatophore in the photosynthetic amoeba *Paulinella*. *Environmental Microbiology Reports*, e13304, 2024. [DOI/URL]

2024 Van Etten J., **Stephens T. G.**, Chille E., Lipzen A., Peterson D., Barry K., Grigoriev I. V., and Bhattacharya D. Diverse fates of ancient horizontal gene transfers in extremophilic red algae. *Environmental Microbiology*, 26(5):e16629, 2024. [DOI/URL]

2024 Benites L. F., **Stephens T. G.**, Van Etten J., James T., Christian W. C., Barry K., Grigoriev I. V., McDermott T. R. & Bhattacharya D. Viruses associated with extremophilic red algal mats reveal signatures of early thermal adaptation. *Communications Biology*, 2024. [DOI/URL] Featured by Science

2023 Bhattacharya D., **Stephens T. G.**, Chille E. E., Benites L. F., Chan C. X. Facultative lifestyle drives diversity of coral algal symbionts. *Trends in Ecology and Evolution*, 39(3):239-247, 2023. [DOI/URL]

2023 *Williams A., ***Stephens T. G.**, Shumaker A., and Bhattacharya D. Peeling back the layers of coral holobiont multi-omics data. *iScience*, 26(9):107623, 2023. [DOI/URL] *Co-first authorship

2023 **Stephens T. G.**, Strand E. L., Putnam H. M., and Bhattacharya D. Ploidy variation and its implications for reproduction and population dynamics in two sympatric Hawaiian coral species. *Genome Biology and Evolution*, evad149, 2023. [Preprint] [DOI/URL]

2023 Van Etten J., **Stephens T. G.**, and Bhattacharya D. A *k*-mer-based approach for inferring phylogenetic relatedness of environmental genomic data. *Systems Biology*, syad037, 2023. [DOI/URL] Won the Publisher's Award for Excellence in Systematic Research

2023 Van Etten J., Benites F. L., **Stephens T. G.**, Yoon H. S., and Bhattacharya D. Algae obscura: The potential of rare species as model systems. *Journal of Phycology*, 59(2):293-300, 2023. [DOI/URL] Ranked within the top 10% of most-viewed papers published by the journal in 2023.

2022 **Stephens T. G.**, Lee J., Jeong Y., Yoon H. S., Putnam H. M., Majerova E., and Bhattacharya D. High-quality genome assemblies from key Hawaiian coral species. *GigaScience*, 11:giac098, 2022. [DOI/URL]

2022 Benites L. F., **Stephens T. G.**, and Bhattacharya D. Multiple waves of viral invasions in Symbiodiniaceae algal genomes. *Virus Evolution*, 8:veac101, 2022. [Preprint] [DOI/URL]

2022 Bhattacharya D., Van Etten J., Benites L. F., and **Stephens T. G.** Endosymbiotic ratchet accelerates divergence after organelle origin. *BioEssays*, e2200165, 2022. [DOI/URL]

2022 *Gabr A., ***Stephens T. G.**, and Bhattacharya D. Loss of key endosymbiont genes may facilitate early host control of the chromatophore in *Paulinella*. *iScience*, 25:104974, 2022. [DOI/URL] *Co-first authorship

2022 Meng Z., Williams A., Liau P., **Stephens T. G.**, Drury C., Chiles E. N., Su X., Javanmard M., and Bhattacharya D. Development of a portable toolkit to diagnose coral thermal stress. *Scientific Reports*, 12:14398, 2022. [DOI/URL]

2022 Bhattacharya D., **Stephens T. G.**, Tinoco A., Richmond R., and Cleves P. A. Life on the edge: Hawaiian model for coral evolution. *Limnology and Oceanography*, 67:1976-1985, 2022. [DOI/URL]

2022 *Calatreva V., ***Stephens T. G.**, Gabr A., Grossman A. R., and Bhattacharya D. Retrotransposition facilitated the establishment of a primary plastid in the thecate amoeba *Paulinella*. *PNAS*, 119:e2121241119, 2022. [DOI/URL] *Co-first authorship

2022 *Gabr A., ***Stephens T. G.**, and Bhattacharya D. Hypothesis: *Trans*-splicing generates evolutionary novelty in the photosynthetic amoeba *Paulinella*. *Journal of Phycology*, 58:392-405, 2022. [DOI/URL] *Co-first authorship

2022 Dougan K. E., Gonzalez-Pech R. A., **Stephens T. G.**, Shah S., Chen Y., Ragan M. A., Bhattacharya D., and Chan C. X. Genome-powered classification of microbial eukaryotes: focus on coral algal symbionts. *Trends in Microbiology*, 30:831-840, 2022. [DOI/URL]

2022 Gabr A., Zournas A., **Stephens T. G.**, Dismukes G., and Bhattacharya D. Evidence for a robust photosystem II in the photosynthetic amoeba *Paulinella*. *New Phytologist*, 234:934-945, 2022. [DOI/URL]

2021 Williams A., Pathmanathan J. S., **Stephens T. G.**, Su X., Chiles E. N., Conetta D., Putnam H. M., and Bhattacharya D. Multi-omic characterization of the thermal stress phenome in the stony coral *Montipora capitata*. *PeerJ*, 9:e12335, 2021. [Preprint] [DOI/URL] Featured by PeerJ Expert Curations: Molecular Ecology: PeerJ Expert Curations, PeerJ Blog

2021 **Stephens T. G.**, Gabr A., Calatreva V., Grossman A. R., and Bhattacharya D. Why is primary endosymbiosis so rare?. *New Phytologist*, 231:1693-1699, 2021. [DOI/URL]

2021 Bernard G., **Stephens T. G.**, Gonzalez-Pech R. A., and Chan C. X. Inferring phylogenomic relationship of microbes using scalable alignment-free methods. *Methods in Molecular Biology*, 2242:69-76, 2021. [DOI/URL]

2021 Jacobus A. P., **Stephens T. G.**, Youssef P., Gonzalez-Pech R., Ciccotosto-Camp M. M., Dougan K. E., Chen Y., Basso L. C., Frazzon J., Chan C. X., and Gross J. Comparative genomics supports that Brazilian bioethanol *Saccharomyces cerevisiae* comprise a unified group of domesticated strains related to cachaca spirit yeasts. *Frontiers in Microbiology*, 12:644089, 2021. [Preprint] [DOI/URL]

2021 Gonzalez-Pech, R. A., **Stephens T. G.**, Chen Y., Mohamed A. R., Cheng Y., Shah S., Dougan K. E., Fortuin M. D. A., Lagorce R., Burt D. W., Bhattacharya D., Ragan M. A., and Chan C. X. Comparison of 15 dinoflagellate genomes reveals extensive sequence and structural divergence in family Symbiodiniaceae and genus *Symbiodinium*. *BMC Biology*, 0.842361111, 2021. [Preprint] [DOI/URL]

2020 Lhee D., Lee J., Ettahi K., Cho C. H., Ha J., Chan Y., Zelzion U., **Stephens T. G.**, Price D. C., Gabr A., Nowack E. C. M., Bhattacharya D., and Yoon H. S. Amoeba genome reveals dominant host contribution to plastid endosymbiosis. *Molecular Biology and Evolution*, 38:344-357, 2020. [DOI/URL]

2020 **Stephens T. G.**, Gonzalez-Pech R. A., Cheng Y., Mohamed A. R., Burt D. W., Bhattacharya D., Ragan M. A., and Chan C. X. Genomes of the dinoflagellate *Polarella glacialis* encode tandemly repeated single-exon genes with adaptive functions. *BMC Biology*, 18:56, 2020. [Preprint] [DOI/URL] Featured by IMB News

2019 Chen Y., Gonzalez-Pech R. A., **Stephens T. G.**, Bhattacharya D., and Chan C. X. Evidence that inconsistent gene prediction can mislead analysis of dinoflagellate genomes. *Journal of Phycology*, 56:6-10, 2019. [Preprint] [DOI/URL]

2018 *Gonzalez-Pech R. A., ***Stephens T. G.**, and Chan C. X. Commonly misunderstood parameters of NCBI BLAST and important considerations for users. *Bioinformatics*, 35:2697-6998, 2018. [DOI/URL] *Co-first authorship [F1000 recommended]

2018 **Stephens T. G.**, Ragan M. A., Bhattacharya D., and Chan C. X. Core genes in diverse dinoflagellate lineages include a wealth of conserved dark genes with unknown functions. *Scientific Reports*, 8:17175, 2018. [DOI/URL]

2018 Liu H., **Stephens T. G.**, Gonzalez-Pech R. A., Beltran V. H., Lapeyre B., Bongaerts P., Cooke I., Aranda M., Bourne D. G., Foret S., Miller D. J., van Oppen M. J. H., Voolstra C. R., Ragan M.A., and Chan C.X. *Symbiodinium* genomes reveal adaptive evolution of functions related to coral-dinoflagellate symbiosis. *Communications Biology*, 0.107638889, 2018. [Preprint] [DOI/URL] Featured by multiple outlets: IMB News, GBRF, Video Feature, IMB 2018 year in review

2018 Lee J, Yang E. C., Graf L., Yang J. H., Qiu H., Zelzion U., Chan C. X., **Stephens T. G.**, Weber A. P. M., Boo G. H., Boo S. M., Kim K. M., Shin Y., Jung M., Lee S. J., Yim H. S., Lee J. Y., Bhattacharya D., and Yoon H. S. Analysis of the draft genome of the red seaweed *Gracilariopsis chorda* provides insights into genome size evolution in Rhodophyta. *Molecular Biology and Evolution*, 35:1869-1886, 2018. [DOI/URL]

2016 **Stephens T. G.**, Bhattacharya D., Ragan M. A., and Chan C. X. PhySortR: a fast, flexible tool for sorting phylogenetic trees in R. *PeerJ*, 4:e2038, 2016. [Preprint] [DOI/URL]

Invited talks and conference presentations

2024 **Stephens T. G.**, Etten J. V., Benites L. F., Mcdermott T., and Bhattacharya D. Community-wide interactions sustain life in geothermal spring habitats. *Rutgers University Microbiome Program Work in Progress Seminar*. Rutgers, USA.

2024 **Stephens T. G.**, Kulczyk A. W., and Bhattacharya D. Cosmopolitan gene families with known functions are hotspots for the evolution of novel genes in stony corals. *3rd Joint Congress on Evolutionary Biology*. 26-30 July 2024, Montreal, Canada.

2023 **Stephens T. G.**, Etten J. V., Benites L. F., Mcdermott T., and Bhattacharya D. Cyanidiophyceae: The extremophilic red algae that underpin hot spring microbial communities in Yellowstone National Park. *UQ Marine Ecogenomics Symposium 2023*. 13-14 December 2023, University of Queensland, Australia.

2023 **Stephens T. G.**, Etten J. V., Benites L. F., Mcdermott T., and Bhattacharya D. Cyanidiophyceae: The extremophilic red algae that underpin hot spring microbial communities in Yellowstone National Park. *Department of Molecular Biology and Biochemistry Seminar*, Rutgers University.

2023 **Stephens T. G.**, Chille E., Wong K., Andrade N., Hulett R., Cleves P., Bhattacharya D., Traylor-Knowles N. Shining a light on "dark" proteins in corals. *From Postdoc to Principal Investigator: An NSF Division of Integrative Organismal Systems (IOS) Virtual Colloquium 2023*. 13-14th September 2023, Virtual.

2023 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Exploring the origin and evolution of primary plastids using *Paulinella* as a model system. *International Society of Endocytobiology conference 2023*. 10-14th September 2023, Field Museum, Chicago.

2023 **Stephens T. G.**, Chille E., Strand E. L., Putnam H. M., and Bhattacharya D. Multi-omics investigation of coral resilience. *Invited talk at Carnegie Plant Biology Seminar, August 2023*. 25th August 2023, Carnegie Department of Plant Biology, Stanford University, USA.

2023 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Exploring the origin and evolution of primary plastids using *Paulinella* as a model system. *New Phytologist Next Generation Scientists Conference 2023*. 3rd July 2023, National University of Singapore, Singapore.

2023 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Exploring endosymbiosis and biotic interactions using niche, non-model algae. *Invited talk at SCMB GenGen/ACE Special Seminar*. 10th January 2023, The University of Queensland, Brisbane, Australia.

2022 **Stephens T. G.**, Williams A., Shumaker A., and Bhattacharya D. Integration of multi-omics coral data under thermal stress . *4th Institute for Food, Nutrition, and Health*. 4th November 2022, Rutgers University, New Jersey, USA.

2022 **Stephens T. G.**, Strand E. L., Putnam H. M., and Bhattacharya D. Differences in ploidy and the prevalence of clonal propagation between *Montipora capitata* and *Pocillopora acuta* from Kane'ohe Bay, Hawai'i. *15th International Coral Reef Symposium*. 3-8th July 2022, Bremen, Germany.

2022 **Stephens T. G.**, Etten J. V., McDermott T., and Bhattacharya D. Analysis of environmental meta-omics data from the extremophilic red algae Cyanidiophyceae. *Joint Aquatic Sciences Meeting*. 14-20th May, 2022, Grand Rapids, USA.

2021 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Insights into the evolution of a primary endosymbiosis through analysis of the *Paulinella* genome. *75th Annual Meeting of the Phycological Society of America*. 13-22nd July 2021, online.

2021 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Insights into the evolution of a primary endosymbiosis through analysis of the *Paulinella* genome. *12th International Phycological Congress*. 22-26th March 2021, Chile.

2019 **Stephens T. G.**, Bhattacharya D., Ragan M. A., and Chan C. X. *Polarella* genomics: understanding the evolutionary transition to algal symbiosis and cold adaptation. *Joint Academic Microbiology Seminars* (*JAMS*). 9th April 2019, Brisbane, Australia.

2018 **Stephens T. G.**, Bhattacharya D., Ragan M. A., and Chan C. X. *Polarella* genomics: understanding the evolutionary transition to algal symbiosis and cold adaptation. *Botany Department, Biosciences Institute, University of Sao Paulo*. 14th December 2018, Sao Paulo, Brazil.

2018 **Stephens T. G.**, Bhattacharya D., Ragan M. A., and Chan C. X. *Polarella* genomics: understanding the evolutionary transition to algal symbiosis and cold adaptation. *2nd Bioenergy Workshop, UNESP-USP-UNICAMP Integrated Postgraduate Program in Bioenergy, Institute for Research in Bioenergy, State University of Sao Paulo* (*UNESP*). 6th December 2018, Sao Paulo, Brazil.

2017 **Stephens T. G.**, Bhattacharya D., Ragan M. A. and Chan C. X. Insights into coral reef symbiosis from the genome of cold-adapted algae. *EMBL Australia Postgraduate Symposium 2017*, 29th November-1st December 2017, Sydney, Australia.

Poster presentations

2024 **Stephens T. G.**, Etten J. V., McDermott T. R., and Bhattacharya D. Cyanidiophyceae: The Extremophilic Red Algae That Underpin Hot Spring Microbial Communities In Yellowstone National Park. *Eastern Regional Photosynthesis Conference 2024*. 12-14th April 2024, Marine Biological Laboratory, Massachusetts, USA.

2024 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Exploring the origin and evolution of primary plastids using *Paulinella* as a model system. *RUMP Annual Retreat 2024*. 16th March 2024, Rutgers University, New Jersey, USA.

2023 **Stephens T. G.**, Chille E., Wong K., Andrade N., Hulett R., Cleves P., Bhattacharya D., Traylor-Knowles N. Shining a light on "dark" proteins in corals. *NSF EDGE PI Meeting 2023*. 10-11th October 2023, Alexandria, Virginia, USA.

2023 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Exploring the origin and evolution of primary plastids using *Paulinella* as a model system. *JGI Annual User Meeting 2023*. 21-24th August 2023, JGI, Berkeley, California, USA.

2018 **Stephens T. G.**, Bhattacharya D., Ragan M. A., and Chan C. X. *Polarella* genomics: understanding cold adaptation and evolutionary transition to symbiosis in dinoflagellates. *Society for Molecular Biology & Evolution* (*SMBE*) *annual meeting 2018*, 8-12th July 2018, Yokohama, Japan.

2016 **Stephens T. G.**, Chan C. X., and Ragan M. *Polarella* genomics: understanding the evolutionary transition to algal symbiosis and cold adaptation, *IMB Research Higher Degree Student Symposium*, 13th Jul 2016, The University of Queensland, Brisbane, Australia.

Selected research coverage

2026 Our paper Crawling under the radar: Two novel *Paulinella* species expand knowledge about the ecology and evolution of a primary plastid?containing amoeba lineage covered by The New York Times

2026 Our Sargassum Biofuel project was featured by Rutgers News

2026 Our paper Gene transfer drives community cooperation in geothermal habitats was featured in an article by Phys.org

2026 Our paper Shifts in the microbiome and virome are associated with stony coral tissue loss disease (SCTLD) was covered by Rutgers Climate and Energy Institute

2025 Our paper Metaproteome Analysis of Short-Term Thermal Stress in Three Sympatric Coral Species Reveals Divergent Host Responses was covered by Rutgers Newsroom

2024 News release by the New Jersey CSIT announcing the recipients of the Clean Tech Seed Grant, which includes our company, OceanOmics.

2024 Our Sargassum Biofuel project was featured by Rutgers News

2024 Our paper Viruses associated with extremophilic red algal mats reveal signatures of early thermal adaptation was featured by Science

2023 Our paper A *k*-mer-Based Approach for Phylogenetic Classification of Taxa in Environmental Genomic Data won the Publisher's Award for Excellence in Systematic Research

2022 Our paper Retrotransposition facilitated the establishment of a primary plastid in the thecate amoeba *Paulinella* covered by Rutgers Research

2022 Article by Rutgers Research on our short film The Coral Holobiont Response to Climate Change which won Best Trailer in the Kiez Berlin Film Festival.

2021 Our paper Why is primary endosymbiosis so rare? was covered by Rutgers Newsroom. We also produced two animated videos: Video 1, Video 2

2020 Our paper Amoeba Genome Reveals Dominant Host Contribution to Plastid Endosymbiosis covered by Rutgers Today

Academic service

2023–present Associate Editor: Symbiotic and Parasitic Protists section, Frontiers in Protistology

Patents and intellectual property

2023 Provisional Patent "Protein Markers of Coral Stress" (5431.1025-000:2023-098) Non-provisional US Patent filed 2024/03/27

2023 Provisional Patent "Coral Rapid Strip Test" (5431.1024-000:2023-097)